

PinkSync.io Hosting Architecture

Provider-Agnostic, Deaf-First, Sovereign Infrastructure

Core Principle

PinkSync sits ABOVE cloud providers — they are client APIs, not partners. This architecture ensures:

- Zero vendor lock-in
 - Fail-over capability
 - Cost optimization
 - Sovereignty maintained
-

Domain Structure & Hosting Map

Root Domain: pinksync.io

<code>pinksync.io</code>
<code>(Marketing Site)</code>
<code>Host: Cloudflare Pages (FREE)</code>
<code>Why: Global CDN, DDoS protection, zero cost</code>

Hosting: Cloudflare Pages (FREE tier)

- Static site generator (Astro, Next.js SSG, or Fresh)
- Auto-deploys from GitHub
- Global edge network
- 500 builds/month free
- Unlimited bandwidth

Content:

- Marketing landing page
- Product overview
- Deaf-first accessibility showcase
- Sign language video demos
- MBTQ ecosystem visualization
- Link to sign up / extension download

GitHub Repo: `pinkycollie/pinksync-marketing`



Identity & Auth Subdomain

deafauth.pinksync.io

```
deafauth.pinksync.io
(DeafAUTH Service)
Host: Deno Deploy (FREE → $10/mo)
Why: Rust PASETO, low latency, serverless
```

Hosting: Deno Deploy

- Free: 100k requests/day, 100 GB/month bandwidth
- Paid: \$10/month for 5M requests/day
- Global edge deployment (35+ regions)
- Zero cold starts
- Native TypeScript/Deno support

Stack:

- Deno runtime with Rust PASETO bindings
- Redis for session storage (Upstash free tier)
- PostgreSQL on Supabase (free tier)

Fallback: If Deno Deploy has issues → Cloudflare Workers

Video Communication Subdomain

vcode.pinksync.io

```
vcode.pinksync.io
(VCode AI Service)
Host: Vercel (FREE → $20/mo)
Why: Edge functions, AI integration, fast
```

Hosting: Vercel (connected to your Vercel account)

- Free: 100 GB bandwidth, 6,000 minutes serverless
- Pro: \$20/month for 1TB bandwidth
- Optimized for Next.js/Node.js
- Edge functions for low-latency AI calls
- Automatic HTTPS

Stack:

- Node.js/Express backend
- Groq AI integration (real-time transcription)
- MediaPipe for ASL detection
- LiveKit for video streams

Alternative: Hetzner VPS if you need long-running WebSocket connections

- €4.51/month for CX22 (2 vCPU, 4GB RAM)
 - Better for persistent connections than serverless
-

Content Transformation Subdomain

pinkflow.pinksync.io

```
pinkflow.pinksync.io
(Real-time Accessibility Engine)
Host: Deno Deploy (FREE → $10/mo)
Why: Fast text transforms, edge compute
```

Hosting: Deno Deploy

- Stateless transformations
- Edge-optimized for low latency
- Pairs well with browser extension

Stack:

- Deno Fresh (zero-bundle React framework)
 - Caption enhancement algorithms
 - Visual contrast optimization
 - ASL-friendly text simplification
-



Event Orchestration Subdomain

events.pinksync.io

```
events.pinksync.io
(Event Orchestrator)
Host: Hetzner VPS (€4.51/mo)
Why: Persistent connections, Redis pub/sub
```

Hosting: Hetzner Dedicated Server (Germany)

- CX22: €4.51/month (2 vCPU, 4GB RAM, 40GB SSD)
- Better control than serverless
- Run Redis + event broker

- Predictable pricing

Stack:

- Deno runtime
- Redis Streams for pub/sub
- NATS.io for distributed messaging
- Docker Compose for orchestration

Why not serverless?

- Event orchestrator needs persistent WebSocket connections
- Serverless has cold starts and connection limits
- Self-hosted gives full control

RAG Engine Subdomain

rag.pinksync.io

```
rag.pinksync.io
(RAG Engine)
Host: Hetzner VPS (€15/mo)
Why: GPU access, vector DB, large models
```

Hosting: Hetzner Cloud CPX31 (€15/month)

- 4 vCPU, 8GB RAM, 160GB SSD
- Enough for Qdrant vector database
- Run Ollama for local LLM inference (sovereign mode)
- Cost-effective GPU via Hetzner GPU cloud if needed

Stack:

- Qdrant (vector database)
- Ollama (local Llama 3.3 models)

- OpenAI embeddings (production)
- Supabase pgvector (backup)

Alternative: Cloudflare R2 for vector storage

- \$0.015 per GB stored
 - No egress fees
 - Pairs well with Cloudflare Workers for queries
-

API Gateway Subdomain

api.pinksync.io

```
api.pinksync.io
(API Gateway)
Host: Cloudflare Workers (FREE)
Why: Routing, rate limiting, DDoS protection
```

Hosting: Cloudflare Workers (FREE tier)

- 100k requests/day free
- Global edge network (300+ locations)
- Built-in DDoS protection
- 10ms CPU time per request

Purpose:

- Route requests to correct microservices
- Authentication middleware
- Rate limiting
- Request logging

Stack:

- Cloudflare Workers (JavaScript/TypeScript)

- Hono.js framework (lightweight routing)
 - JWT validation
 - Request forwarding to backend services
-

Studio Subdomain

studio.pinksync.io

```
studio.pinksync.io
(Creator Studio UI)
Host: Vercel (FREE → $20/mo)
Why: React app, fast deployments, preview URLs
```

Hosting: Vercel

- Frontend for SignMirror creator marketplace
- Next.js React app
- Real-time collaboration features
- Preview deployments for every PR

Stack:

- Next.js 15 (App Router)
 - Supabase Realtime for collaboration
 - FibonRose trust sync
 - Three.js for 3D avatar rendering
-

Dashboard Subdomain

dash.pinksync.io

```
dash.pinksync.io
```

(Admin Dashboard)
Host: Cloudflare Pages (FREE)
Why: Static dashboard, API calls to backends

Hosting: Cloudflare Pages

- SvelteKit or Astro static site
 - Connects to API via `api.pinksync.io`
 - User management
 - Service monitoring
 - Analytics
-

Blockchain Trust Subdomain

trust.pinksync.io

trust.pinksync.io
(FibonRose Logging)
Host: Hetzner VPS (€4.51/mo)
Why: Blockchain writes, persistent state

Hosting: Hetzner VPS CX22

- Long-running blockchain sync process
- Thirdweb SDK integration
- IPFS pinning for trust logs
- PostgreSQL for indexed data

Stack:

- Deno runtime
- Thirdweb SDK
- Base blockchain (low gas fees)

- IPFS via Pinata (free tier: 1GB)
-

Search & Discovery Subdomain

search.pinksync.io

```
search.pinksync.io
(Semantic Search API)
Host: Deno Deploy (FREE → $10/mo)
Why: Fast vector search queries
```

Hosting: Deno Deploy

- Queries RAG engine
 - Connects to Qdrant
 - Handles semantic search
 - Returns ASL-friendly results
-

Documentation Subdomain

docs.pinksync.io

```
docs.pinksync.io
(Technical Docs)
Host: GitHub Pages (FREE)
Why: Simple, versioned, zero cost
```

Hosting: GitHub Pages (FREE)

- MkDocs or Docusaurus
- Auto-deploys from `docs/` folder
- Version control built-in

- WCAG AAA compliant

Alternative: Cloudflare Pages (same benefits + better performance)

Extension Store Subdomain

extension.pinksync.io

```
extension.pinksync.io
(Extension Download Page)
Host: Cloudflare Pages (FREE)
Why: Static page, fast CDN delivery
```

Hosting: Cloudflare Pages

- Landing page for extension
 - Direct links to Chrome Web Store
 - Firefox Add-ons
 - Edge Add-ons
 - Installation guides
-

Database Strategy

Primary Database: Supabase (FREE → \$25/mo)

db.pinksync.io (CNAME to Supabase)

Free Tier:

- 500 MB database
- 1 GB file storage
- 2 GB bandwidth
- 50k monthly active users

Pro Tier (\$25/month):

- 8 GB database
- 100 GB file storage
- 250 GB bandwidth

Use for:

- User accounts (DeafAUTH)
 - Preferences
 - VCode transcripts
 - Scaffold metadata
-

Alternative: Self-Hosted on Hetzner

```
PostgreSQL on Hetzner VPS (€15/mo)  
For full sovereignty
```

If you need complete control:

- Hetzner CPX31 (€15/month)
 - PostgreSQL 15
 - Daily backups to Hetzner Object Storage
 - pgvector extension for RAG
-

Object Storage Strategy

Option 1: Cloudflare R2 (RECOMMENDED)

`cdn.pinksync.io` → Cloudflare R2

Pricing:

- \$0.015 per GB stored per month

- **Zero egress fees** (huge savings)
- S3-compatible API

Use for:

- Generated scaffolds (artifacts)
- User uploads
- ASL video library
- Extension downloads

Why R2 over S3?

- At 10TB scale: S3 costs ~\$900/month in egress
 - R2 costs: \$150/month (storage only)
 - **\$750/month savings**
-

Option 2: AWS S3 (System of Record)

`s3.pinksync.io` → AWS S3

Keep as canonical storage:

- Deep integration with existing infrastructure
- Lifecycle policies
- Glacier for archival
- Pair with CloudFront for CDN

Cost Optimization:

- Use S3 for writes/archival
 - Mirror to R2 for reads
 - 10x cheaper egress
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Static Assets: Cloudflare CDN (FREE)

`static.pinksync.io` → Cloudflare

Assets:

- JavaScript bundles
- CSS files
- Images
- Fonts
- Icons

Free Tier:

- Unlimited bandwidth
 - Global caching
 - DDoS protection
 - Automatic HTTPS
-



SSL/TLS Strategy

All subdomains: Cloudflare SSL (FREE)

Cloudflare provides:

- Free Universal SSL
- Automatic certificate renewal
- Full (strict) encryption mode
- TLS 1.3

Setup:

1. Point `pinksync.io` nameservers to Cloudflare
2. All subdomains get automatic SSL
3. Zero configuration needed

DNS Configuration

Move DNS to Cloudflare (if not already)

Nameservers:

```
ava.ns.cloudflare.com
bob.ns.cloudflare.com
```

DNS Records:

```
# Root domain
pinksync.io          A          <Cloudflare Pages IP>
www.pinksync.io      CNAME    pinksync.io

# API Gateway (Cloudflare Workers)
api.pinksync.io      CNAME    api.workers.dev

# Microservices (Deno Deploy)
deafauth.pinksync.io CNAME    pinksync-deafauth.deno.dev
pinkflow.pinksync.io CNAME    pinksync-pinkflow.deno.dev
search.pinksync.io   CNAME    pinksync-search.deno.dev

# Vercel-hosted
vcode.pinksync.io    CNAME    cname.vercel-dns.com
studio.pinksync.io   CNAME    cname.vercel-dns.com

# Hetzner VPS (get IP from Hetzner panel)
events.pinksync.io   A          <Hetzner VPS IP>
rag.pinksync.io       A          <Hetzner VPS IP>
trust.pinksync.io    A          <Hetzner VPS IP>

# Static sites
docs.pinksync.io     CNAME    pinkycollie.github.io
dash.pinksync.io     CNAME    dash-pinksync.pages.dev
extension.pinksync.io CNAME    extension-pinksync.pages.dev

# Database & Storage
db.pinksync.io        CNAME    <supabase-project>.supabase.co
cdn.pinksync.io       CNAME    <r2-bucket>.r2.cloudflarestorage.com
```

Total Cost Breakdown

Minimal Configuration (Solo Founder Budget)

Service	Host	Monthly Cost
<code>pinksync.io</code>	Cloudflare Pages	\$0
<code>deafauth.pinksync.io</code>	Deno Deploy	\$0 (free tier)
<code>pinkflow.pinksync.io</code>	Deno Deploy	\$0 (free tier)
<code>vcode.pinksync.io</code>	Vercel	\$0 (free tier)
<code>api.pinksync.io</code>	Cloudflare Workers	\$0
<code>events.pinksync.io</code>	Hetzner CX22	€4.51
<code>rag.pinksync.io</code>	Hetzner CPX31	€15.00
<code>trust.pinksync.io</code>	(same VPS)	€0
<code>docs.pinksync.io</code>	GitHub Pages	\$0
Database	Supabase	\$0 (free tier)
Storage	Cloudflare R2	~\$5 (100GB)
DNS & SSL	Cloudflare	\$0
TOTAL		~€25/month

Production Scale (Post-Funding)

Service	Host	Monthly Cost
<code>pinksync.io</code>	Cloudflare Pages	\$0
<code>deafauth.pinksync.io</code>	Deno Deploy Pro	\$10
<code>pinkflow.pinksync.io</code>	Deno Deploy Pro	\$10
<code>vcode.pinksync.io</code>	Vercel Pro	\$20
<code>api.pinksync.io</code>	Cloudflare Workers Paid	\$5

events.pinksync.io	Hetzner CCX23	€31.00
rag.pinksync.io	Hetzner CPX51	€49.00
trust.pinksync.io	(same VPS)	€0
Database	Supabase Pro	\$25
Storage	Cloudflare R2	\$50 (1TB)
Monitoring	Grafana Cloud	\$8
TOTAL		~€180/month

Deployment Workflow

1. Development Flow

```
# Developer pushes code
git push origin feat-caption-enhancement

# GitHub Actions triggers
→ Runs tests
→ Builds service
→ Deploys to preview URL
→ Creates staging deployment
```

2. Production Flow

```
# Merge to main branch
git checkout main
git merge feat-caption-enhancement
git push origin main

# Auto-deployment sequence:
1. Cloudflare Pages: pinksync.io updates
2. Deno Deploy: deaafauth/pinkflow deploy
3. Vercel: vcode/studio deploy
4. Hetzner: SSH + Docker Compose pull
5. Health checks run
6. Slack notification sent
```

Failover Strategy

Service Redundancy

Each critical service has a backup:

Primary: Deno Deploy
Fallback: Cloudflare Workers

Primary: Vercel
Fallback: Hetzner VPS

Primary: Supabase
Fallback: Self-hosted PostgreSQL on Hetzner

Load Balancer Configuration

```
# Nginx on Hetzner VPS for critical services

upstream deafauth {
    server deafauth-primary.deno.dev;
    server deafauth-backup.workers.dev backup;
}

upstream vcode {
    server vcode-primary.vercel.app;
    server vcode-backup.hetzner.local backup;
}
```

Why This Architecture?

Sovereignty Maintained

- No single vendor lock-in
- Can migrate services independently
- Full control over critical infrastructure

Cost Optimized

- Free tiers for low-traffic services
- Hetzner for predictable costs
- R2 eliminates egress fees

Performance

- Edge compute (Deno, Cloudflare) for low latency
- Regional VPS (Hetzner) for persistent connections
- CDN for global asset delivery

Deaf-First

- Fast caption processing (edge compute)
- Low-latency video (Vercel edge)
- Accessible everywhere (global CDN)

Grant-Ready

- Clear infrastructure documentation
- Scalable architecture
- Security best practices
- Accessibility compliance (WCAG AAA)



Implementation Checklist

Phase 1: DNS & Domain Setup (Week 1)

- Move DNS to Cloudflare
- Configure all CNAME records
- Enable SSL for all subdomains
- Test DNS propagation

Phase 2: Core Services (Week 2)

- Deploy DeafAUTH to Deno Deploy

- Deploy PinkFlow to Deno Deploy
- Deploy VCode to Vercel
- Set up API Gateway on Cloudflare Workers

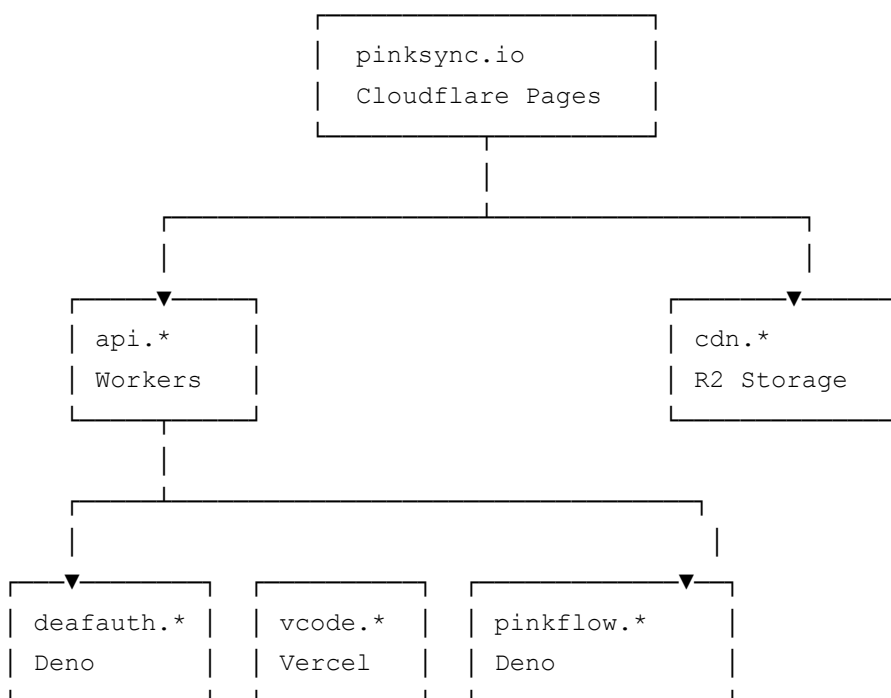
Phase 3: Infrastructure (Week 3)

- Provision Hetzner VPS for event orchestrator
- Set up PostgreSQL on Supabase
- Configure Cloudflare R2 buckets
- Deploy RAG engine to Hetzner

Phase 4: Monitoring (Week 4)

- Set up Grafana Cloud
- Configure health checks
- Connect Slack notifications
- Test failover scenarios

Final Architecture Visualization



🌟 **This is the nervous system you envisioned.**

Every subdomain has a purpose. Every service has a home. The ecosystem is sovereign, scalable, and Deaf-first.

Ready to deploy? I can help you configure any specific subdomain next.